

# Package #159

deliberately introduce a thin, reactive silane-based interlayer that forms a strong covalent bond with the glass surface, consuming mobile species at the interface and “pinning” adjacent grain boundaries, thus converting the originally active surface into a passive grain-boundary-locking layer.

**Technology Maturity:** ENGINEERING\_DEFINITION

**Transfer Posture:** Sponsored Validation

**Problem:** Documented problem 'grain boundary sliding' concerning borosilicate glass reactor liner. User-submitted engineering problem (session input): "Why does grain boundary sliding in borosilicate glass reactor liners limit their service temperature, and can a grain-boundary engineering treatment prevent grain boundary sliding in the glass liner?". [cod\_optimade] retrieval status EMPTY — provider outcome recorded, not treated as absence (Art. XXI.3). [arxiv] 5 records retrieved live (1903-2021). Illustrative records: 'Motion of grain boundaries incorporating dislocation structure'; 'Energy of low angle grain boundaries based on continuum dislocation structure'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

**Mechanism:** the surface of common laboratory borosilicate glassware is, incorrectly, assumed to be chemically inert.

## Contents:

- 01 - Executive Technology Brief
- 02 - Engineering Technology-Transfer Dossier
- 03 - Buyer Decision Card
- 04 - Evidence Summary
- 05 - Transfer Manifest

*Status: ENGINEERING\_DEFINITION. Not physically validated.*